

AI ExamGuard — Framework & Library Reference

Every third-party dependency actually imported by the codebase, mapped to the module that uses it and what it does there.

Generated 2026-08-12 · Backend: FastAPI/Python · Frontend: React/Vite · Extension: Chrome MV3

Backend — Web Framework & API Layer

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
FastAPI	0.139.0	backend/app/main.py backend/app/routes/*.py	Core web framework. Defines every REST endpoint (auth, schools, courses, exams, sessions, violations...), registers routers, and drives dependency-injected auth guards (Depends(get_current_user), require_admin, require_instructor) on each route.
Starlette	1.3.1	backend/app/main.py	The ASGI toolkit FastAPI is built on. Supplies CORSMiddleware, which restricts which frontend origins may call the API (ALLOWED_ORIGINS).
Uvicorn	0.51.0	process entrypoint	ASGI server that actually runs the FastAPI app (uvicorn app.main:app), both in local dev and inside the Docker container.
Pydantic / pydantic-settings	2.13.4 / 2.14.2	backend/app/schemas/*.py backend/app/core/config.py	Request/response validation and serialization for every endpoint (ExamCreate, StudentResponse, etc.), plus typed environment-variable loading (DATABASE_URL, SECRET_KEY) via Settings.
python-multipart	0.0.32	backend/app/routes/face.py, object_detection.py, violation.py	Lets FastAPI parse multipart/form-data — required for every file upload: webcam frames, evidence screenshots, CSV question imports.
email-validator	2.3.0	backend/app/schemas/*.py	Backs Pydantic's EmailStr field type used on registration/user schemas so malformed emails are rejected at the API boundary.
python-dotenv	1.2.2	backend/app/core/config.py	Loads backend/.env (DB URL, JWT secret, CORS origins) into the process environment at startup.

Backend — Database & Persistence

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
SQLAlchemy	2.0.51	backend/app/models/*.py backend/app/services/*.py	ORM layer. Every table (User, Exam, ExamSession, Violation, ExamRoster...) is a mapped class here; all service-layer queries and relationships (cascades, joins for multi-tenant scoping) run through it.
Alembic	1.18.5	backend/alembic/versions/*.py	Schema migration tool. Every DB change (new column, table, constraint) is a versioned migration script; runs automatically on every container start per the deployment runbook.
psycopg2-binary	2.9.12	used via SQLAlchemy engine	The PostgreSQL DBAPI driver — the actual wire-protocol connection SQLAlchemy uses to talk to the Postgres 18 container.

Backend — Auth & Security

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
passlib + bcrypt	1.7.4 / 4.0.1	backend/app/auth/security.py	CryptContext(schemes=["bcrypt"]) hashes passwords on register and verifies them on login. Version-pinned deliberately — bcrypt above 4.0.1 breaks this passlib version (see repo CLAUDE.md).
python-jose	3.5.0	backend/app/auth/jwt.py	jwt.encode/jwt.decode issue and validate the JWT access token returned by POST /auth/login and read on every authenticated request by get_current_user.
cryptography, rsa, ecdsa, pyasn1	—	transitive	Cryptographic primitives that python-jose's signing backend depends on — not imported directly by app code.

Backend — Computer Vision & Proctoring ML

The real, trained-model core of the product — no mocked ML, per project convention.

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
opencv-contrib-python	5.0.0.93	backend/app/services/face_service.py	cv2.FaceDetectorYN_create (YuNet) locates a face in the webcam frame; cv2.face.LBPHFaceRecognizer_create matches it against the student's enrolled face model for identity verification; cv2.solvePnP estimates head pose (pitch) to detect sustained "head-down" behavior.
ultralytics (YOLOv8)	8.4.104	backend/app/services/object_detection_service.py	YOLO(...) loads a base model plus a fine-tuned phone-detection model and a pose model, run per exam-frame poll to flag a visible phone or more than one person in frame.
torch / torchvision	2.13.0+cu126 / 0.28.0+cu126	underlies ultralytics inference	The tensor/inference engine YOLOv8 actually runs on — GPU-accelerated (CUDA) in local dev, deliberately CPU-only in the deployed Docker image (no GPU passthrough, per CLAUDE.md).
scikit-learn	1.9.0	backend/app/services/risk_model_service.py backend/training/train_risk_model.py	A fitted logistic-regression model turns FACE_LOST/PHONE_DETECTED/MULTIPLE_PEOPLE counts into a vision risk score. Trained offline on the MSU OEP dataset ground truth rather than hand-guessed weights (unlike the behavioral signals in risk_service.py, which are still a fixed weight dict).
joblib	1.5.3	backend/app/services/risk_model_service.py	Deserializes the persisted risk_model.joblib file into the in-memory scikit-learn model at process startup.
numpy	2.5.1	face_service.py, object_detection_service.py	Decodes uploaded image bytes into the array format OpenCV/YOLO expect, and backs most of the numeric image-processing glue between them.
Pillow	12.3.0	image-handling utilities	General image format conversion/support alongside the OpenCV and YOLO pipelines.

Backend — Offline Training & Evaluation Tooling

Used only by scripts in backend/training/ — dataset prep, model fitting, threshold sweeps, fairness/eval audits. Not on the live request path.

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
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<code>pandas</code>	3.0.5	<code>backend/training/*.py</code>	Tabular wrangling of annotation batches, labeled datasets, and frozen-holdout eval results.
<code>polars</code>	1.43.0	<code>backend/training/*.py</code>	Faster dataframe operations for larger training-set processing scripts, used alongside <code>pandas</code> .
<code>matplotlib</code>	3.11.1	<code>backend/training/*.py</code>	Plots eval curves and threshold-sweep charts (e.g. the recall-regression investigation that reverted the face-match threshold from 0.70 back to 0.35).
<code>scipy</code>	1.18.0	<code>backend/training/*.py</code>	Statistical routines supporting the scikit-learn fitting/evaluation scripts (e.g. LOSO cross-validation).

Backend — Testing

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
<code>pytest / pytest-cov / coverage</code>	9.1.1 / 7.1.0 / 7.15.2	<code>backend/tests/*.py</code>	Test runner and coverage reporting for the service/route test suite.

Frontend — Core Framework & Build

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
<code>React / react-dom</code>	19.2.7	<code>frontend/src/**/*.jsx</code>	Component framework for the entire UI — every page (Dashboard variants, ExamRoom, Reports...) and every reusable piece (Card, RiskPill, ConfirmDialog).
<code>Vite</code>	8.1.1	<code>frontend/vite.config.js</code>	Dev server (HMR) and production bundler. <code>vite build</code> produces the static assets served in deployment.
<code>@vitejs/plugin-react</code>	6.0.3	<code>frontend/vite.config.js</code>	Enables JSX transform and React Fast Refresh inside Vite.
<code>react-router-dom</code>	7.18.1	<code>frontend/src/App.jsx</code> <code>frontend/src/components/ProtectedRoute.jsx</code>	Client-side routing. Nests every route under <code>/:schoolSlug</code> for multi-tenancy, and <code>ProtectedRoute</code> gates access by auth state and <code>role_name</code> .

Frontend — Data, UI & Visualization

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
<code>axios</code>	1.18.1	<code>frontend/src/api/*.js</code>	HTTP client wrapping every backend call — auth, exams, exam sessions, violations, face/object checks — attaching the JWT and handling responses for the whole app.
<code>recharts</code>	3.10.0	<code>StudentDashboard.jsx</code> , <code>InstructorDashboard.jsx</code>	Renders the student's risk-over-time area chart and the instructor's live violation-breakdown pie chart.
<code>lucide-react</code>	1.25.0	<code>Dashboard/*</code> , <code>ui/Card.jsx</code> , etc.	Icon set used throughout the UI (BookOpen, Eye, UserCheck, RefreshCw, Trash2...).
<code>Tailwind CSS</code>	4.3.3	every <code>.jsx</code> file's <code>className</code> props	Utility-class styling framework — all layout, color, and typography across the app is written as Tailwind classes rather than separate CSS files.
<code>@tailwindcss/vite</code>	4.3.3	<code>frontend/vite.config.js</code>	Integrates Tailwind's CSS engine directly into the Vite build pipeline.

Frontend — Dev Tooling

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
<code>eslint</code>	1.71.0	<code>pnpm lint</code>	Fast Rust-based linter for the JS/JSX source — no runtime role, dev-time only.
<code>@types/react</code> , <code>@types/react-dom</code>	19.2.x	<code>editor tooling</code>	Type definitions for editor autocomplete/type-checking only — the project itself is plain <code>.jsx</code> , not TypeScript.

Browser Extension — Tab Monitor

LIBRARY	VERSION	MODULE / FILE	ROLE & FUNCTION
— none —	–	extension/background.js extension/manifest.json	Deliberately zero third-party dependencies. Manifest V3 service worker using only native Chrome APIs: <code>chrome.tabs/chrome.webNavigation</code> to watch for AI-tool and search-engine domains in other tabs, and <code>chrome.runtime.connect</code> port messaging to report detections to the exam page (origin-checked against <code>EXPECTED_ORIGIN</code>).

Scope note: this reference lists libraries actually imported by application/extension code, plus the migration and training tooling that ships with the repo. Purely transitive sub-dependencies of the above (e.g. `h11`, `anyio`, `click`, `colorama`) are omitted — they're pulled in by `FastAPI/Uvicorn/python-jose` themselves and never imported directly.